INTAKE-ARBITER -- FREE-COOLING DECISION REPORT

An agent that decides, hour by hour, whether outside air can cool a data centre.

THE SITE

Location	Switch Edge 1, NV -- station KLAS, America/Los_Angeles
Committed pair	OSM 585998678 -> 617194689 Switch Las Vegas 9 / Switch Las Vegas 11
Facade gap	189.3 m between the two halls
Weather record	43,768 real hourly records from KLAS
Plume physics	576 steady-state solves on this site's own footprints; worst intake rise 0.8564 C at 265 deg

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day	2023-03-10 (recirc_edge)
Plant limit	21.0 C
Notice required	3 h
Forecast skill	0.50 relative to persistence
Level anchor	sensor
Condenser bank	longest facade
Switch budget	2 mode changes per day
Minimum dwell	3 h
Max dew point	15.0 C (Green Grid WP#46 p.6)

WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 3 of 24 hours with 2 mode changes. The reactive incumbent took 8 hours -- 5 more -- but declared 0 unsafe hours against the agent's 0. 6 hours were safe but still ran chillers, because the schedule could not afford them.

Agent	3 of 24 hours free cooling, 2 mode change(s), 0 unsafe hour(s)
Incumbent	8 of 24 hours free cooling, 2 mode change(s), 0 unsafe hour(s)
	6 hour(s) passed every gate and still ran chillers, because the schedule could not afford them

EVERY HOUR, AND THE REASON FOR IT

hh	mode	safe	binding	bound	limit	actual	margin
00	mechanical	yes	-	16.960	21.0	17.220	0.881
01	mechanical	yes	-	16.965	21.0	17.220	0.782
02	mechanical	yes	-	16.409	21.0	16.110	0.716
03	mechanical	no	refusal	17.009	21.0	17.220	0.672
04	FREE	yes	-	16.961	21.0	17.220	0.533
05	FREE	yes	-	16.686	21.0	17.220	0.657
06	FREE	yes	-	18.094	21.0	17.780	0.788
07	mechanical	no	refusal	19.502	21.0	18.330	0.972
08	mechanical	yes	-	20.857	21.0	18.890	1.006
09	mechanical	no	dry-bulb	22.250	21.0	20.000	0.978
10	mechanical	no	refusal	22.272	21.0	19.440	1.054
11	mechanical	no	refusal	22.270	21.0	20.000	0.818

12	mechanical	no	refusal	23.190	21.0	21.110	0.843
13	mechanical	no	refusal	22.540	21.0	21.116	0.867
14	mechanical	no	dry-bulb	22.251	21.0	20.564	0.932
15	mechanical	no	dry-bulb	21.414	21.0	18.893	0.828
16	mechanical	no	refusal	20.337	21.0	17.780	0.844
17	mechanical	no	refusal	19.215	21.0	17.220	0.806
18	mechanical	no	refusal	18.346	21.0	17.221	1.025
19	mechanical	no	refusal	17.003	21.0	17.220	0.807
20	mechanical	no	refusal	16.156	21.0	16.670	0.706
21	mechanical	yes	-	16.186	21.0	16.670	0.746
22	mechanical	yes	-	16.461	21.0	16.670	0.916
23	mechanical	no	refusal	15.908	21.0	16.116	0.762

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin
actual = what the intake temperature turned out to be, from the station record
margin = group-conditional forecast error for that hour of day, plus plume spread

THE REASONING, HOUR BY HOUR

00:00 MECHANICAL

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 16.960 C against a 21.0 C limit. The schedule is what forbids it: no relaxation of either the switch budget or the dwell would free it, so the surrounding hours are worth more. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

01:00 MECHANICAL

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 16.965 C against a 21.0 C limit. The schedule is what forbids it: no relaxation of either the switch budget or the dwell would free it, so the surrounding hours are worth more. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

02:00 MECHANICAL

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 16.408 C against a 21.0 C limit. The schedule is what forbids it: no relaxation of either the switch budget or the dwell would free it, so the surrounding hours are worth more. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

03:00 MECHANICAL [refusal]

Mechanical, and the reason is a REFUSAL rather than a temperature. At this wind bearing a building sits between the condensers and the air intake. The dispersion model has no representation of a building standing in the flow, so any number it produced here would be meaningless. The agent declines to certify the hour instead of returning a figure it cannot stand behind.

04:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 16.961 C, which is 4.039 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.533 C, and the margin is measured, not chosen: 0.517 C of group-conditional forecast error for this hour of day, plus 0.0161 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

05:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 16.686 C, which is 4.314 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.657 C, and the margin is measured, not chosen: 0.637 C of group-conditional forecast error for this hour of day, plus 0.0207 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

06:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 18.094 C, which is 2.906 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.788 C, and the margin is measured, not chosen: 0.748 C of group-conditional forecast error for this hour of day, plus 0.0393 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

07:00 MECHANICAL [refusal]

Mechanical, and the reason is a REFUSAL rather than a temperature. At this wind bearing a building sits between the condensers and the air intake. The dispersion model has no representation of a building standing in the flow, so any number it produced here would be meaningless. The agent declines to certify the hour instead of returning a figure it cannot stand behind.

08:00 MECHANICAL

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 20.857 C against a 21.0 C limit. The schedule is what forbids it: no relaxation of either the switch budget or the dwell would free it, so the surrounding hours are worth more. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

09:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 22.250 C against a 21.0 C limit, so it fails by 1.250 C. It would take a limit 1.250 C higher, or a bound 1.250 C tighter, to change this hour.

10:00 MECHANICAL [refusal]

Mechanical, and the reason is a REFUSAL rather than a temperature. At this wind bearing a building sits between the condensers and the air intake. The dispersion model has no representation of a building standing in the flow, so any number it produced here would be meaningless. The agent declines to certify the hour instead of returning a figure it cannot stand behind.

11:00 MECHANICAL [refusal]

Mechanical, and the reason is a REFUSAL rather than a temperature. At this wind bearing a building sits between the condensers and the air intake. The dispersion model has no representation of a building standing in the flow, so any number it produced here would be meaningless. The agent declines to certify the hour instead of returning a figure it cannot stand behind.

12:00 MECHANICAL [refusal]

Mechanical, and the reason is a REFUSAL rather than a temperature. At this wind bearing a building sits between the condensers and the air intake. The dispersion model has no representation of a building standing in the flow, so any number it produced here would be meaningless. The agent declines to certify the hour instead of returning a figure it cannot stand behind.

13:00 MECHANICAL [refusal]

Mechanical, and the reason is a REFUSAL rather than a temperature. At this wind bearing a building sits between the condensers and the air intake. The dispersion model has no representation of a building standing in the flow, so any number it produced here would be meaningless. The agent declines to certify the hour instead of returning a figure it cannot stand behind.

14:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 22.251 C against a 21.0 C limit, so it fails by 1.251 C. It would take a limit 1.251 C higher, or a bound 1.251 C tighter, to change this hour.

15:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 21.414 C against a 21.0 C limit, so it fails by 0.414 C. It would take a limit 0.414 C higher, or a bound 0.414 C tighter, to change this hour.

16:00 MECHANICAL [refusal]

Mechanical, and the reason is a REFUSAL rather than a temperature. At this wind bearing a building sits between the condensers and the air intake. The dispersion model has no representation of a building standing in the flow, so any number it produced here would be meaningless. The agent declines to certify the hour instead of returning a figure it cannot stand behind.

17:00 MECHANICAL [refusal]

Mechanical, and the reason is a REFUSAL rather than a temperature. At this wind bearing a building sits between the condensers and the air intake. The dispersion model has no representation of a building standing in the flow, so any number it produced here would be meaningless. The agent declines to certify the hour instead of returning a figure it cannot stand behind.

18:00 MECHANICAL [refusal]

Mechanical, and the reason is a REFUSAL rather than a temperature. At this wind bearing a building sits between the condensers and the air intake. The dispersion model has no representation of a building standing in the flow, so any number it produced here would be meaningless. The agent declines to certify the hour instead of returning a figure it cannot stand behind.

19:00 MECHANICAL [refusal]

Mechanical, and the reason is a REFUSAL rather than a temperature. At this wind bearing a building sits between the condensers and the air intake. The dispersion model has no representation of a building standing in the flow, so any number it produced here would be meaningless. The agent declines to certify the hour instead of returning a figure it cannot stand behind.

20:00 MECHANICAL [refusal]

Mechanical, and the reason is a REFUSAL rather than a temperature. At this wind bearing a building sits between the condensers and the air intake. The dispersion model has no representation of a building standing in the flow, so any number it produced here would be meaningless. The agent declines to certify the hour instead of returning a figure it cannot stand behind.

21:00 MECHANICAL

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 16.186 C against a 21.0 C limit. The schedule is what forbids it: no relaxation of either the switch budget or the dwell would free it, so the surrounding hours are worth more. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

22:00 MECHANICAL

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 16.461 C against a 21.0 C limit. The schedule is what forbids it: no relaxation of either the switch budget or the dwell would free it, so the surrounding hours are worth more. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

23:00 MECHANICAL [refusal]

Mechanical, and the reason is a REFUSAL rather than a temperature. At this wind bearing a building sits between the condensers and the air intake. The dispersion model has no representation of a building standing in the flow, so any number it produced here would be meaningless. The agent declines to certify the hour instead of returning a figure it cannot stand behind.

Generated by INTAKE-ARBITER/src/report.py. Verified by being read back: the file was reopened after writing and every hour of the schedule, the headline counts and this site's own name were confirmed present.